Chapter 3: Continuous Random Variables

Math 217: Probability & Statistics

This chapter will introduce

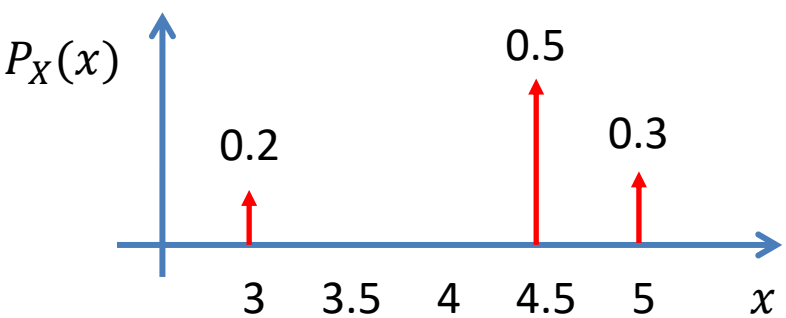
- PDF: Probability Density Function (use in place of PMF)
- CDF: Cumulative Distribution Function
- Expectation and Variance
- Common Cont. RVs

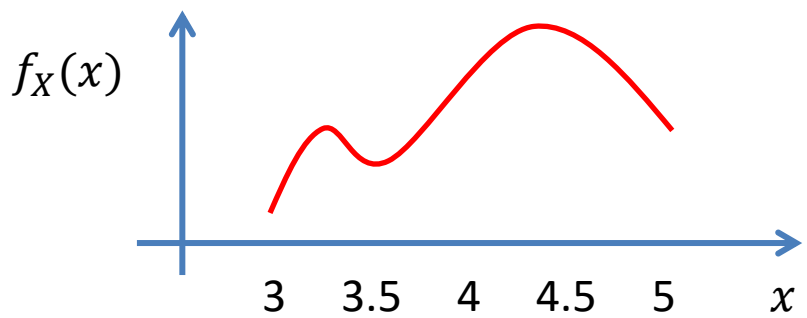
Discrete vs Continuous

- A discrete random variable is a countable set of numbers
  - Rolling a die

- A continuous random variable contains all real numbers between two limits
  - the time for a train you are waiting
  - the heights of all people over 21 age

	Density	**Cumulative**
Discrete R.V.	PMF	CDF
Countinous R.V.	PDF	





PDF-Probability Density Function

Probability **D**ensity **F**unctions can be used to determine the probability that a continuous random variable lies between two values.

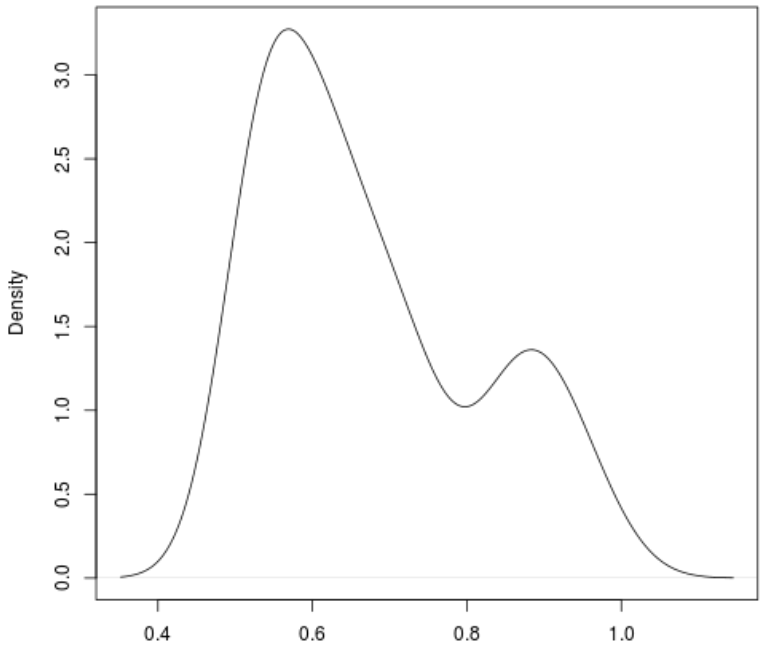
Definition: Probability density function of a cont. random variable can be found by derivative of CDF

Properties of PDF

For a cont R.V. X with $f_X(x)$

- For any interval in sample space , $f_X(x) \geq 0$
- $\int_{-\infty}^{\infty} f_X(x)dx$=1
- $F_X(x) = \int_{-\infty}^{x} f_X(x)dx$

$$f_X(x) = {dF_X(x)}/{dx} \qquad F_X(x) = \int_{-\infty}^{x} f_X(x)d_x$$

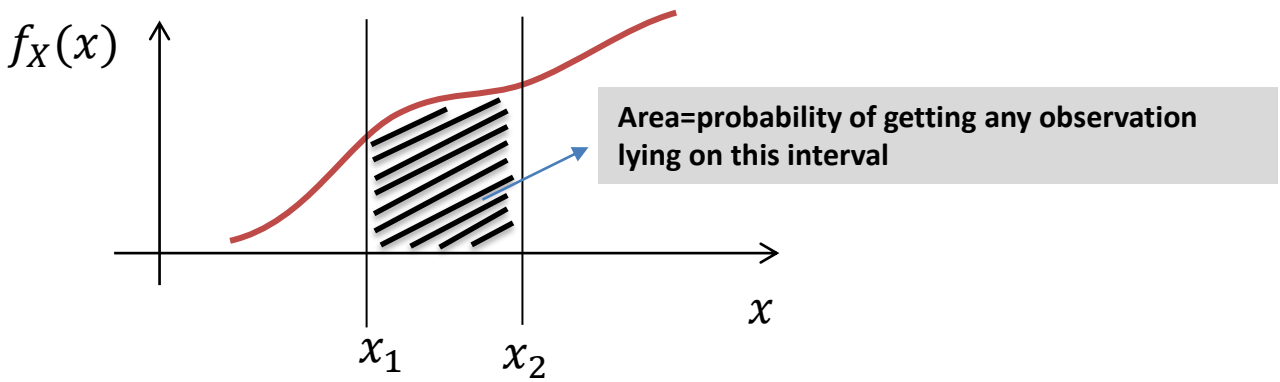


Sample PDF

Theorem 3.1

The probability of interval

$$P(x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} f_X(x)dx$$



Discrete RV

- A **discrete random variable** X takes distinct values in a given interval of numbers.
- The probability distribution of a discrete random variable is shown by a **dirac function**.
- The probability that X is between an interval of numbers is the sum of **diracs**.

Countinous RV

- A **continuous random variable** X takes all values in a given interval of numbers.
- The probability distribution of a continuous random variable is shown by a **density curve**.
- The probability that X is between an interval of numbers is the area under the density curve between the interval end points

Example-3.1

Let X is countinous random variable with probability density function

$$f_x(x) = \begin{cases} 3x^2 & 0 < x \le 1 \\ 0 & otherwise \end{cases}$$

Find the following probabilities

a) P(1/2<X ≤1)
b) P(X=1/2)

$P(1/2 \le X \le 1) = \underline{\underline{0.874}}$

$P(1/2 < X \le 1) = \int_{1/2}^{1} 3x^2 dx = 0.874$

$P(X = 1/2) \Rightarrow \int_{1/2}^{1/2} 3x^2 dx = 0$

$X = 1/2 \rightarrow 1/\infty = 0$

Example-3.2

Let X ~~is~~ be countinous random variable whose probability density function is given by

$$f_X(x) = \begin{cases} C(4x - 2x^2) & 0 < x < 2 \\ 0 & otherwise \end{cases}$$

$$\int_0^2 C(4x-2x^2)\,dx = 1$$

$$C = 3/8$$

a) What is the value of C
b) Find P(X>1)

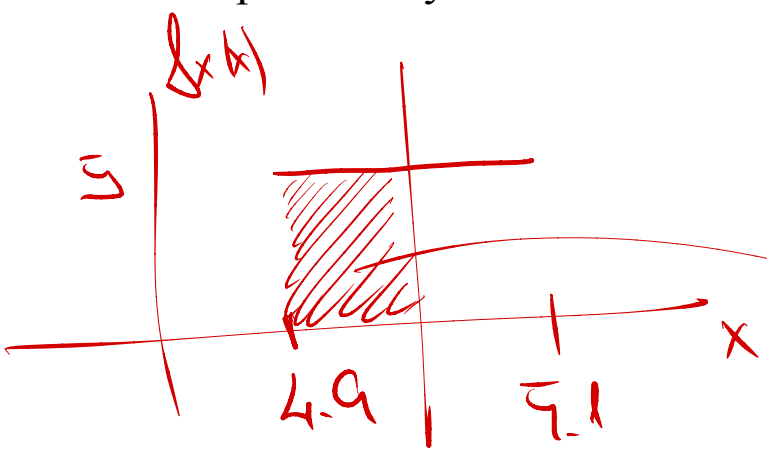
$$P(X>1) = \int_1^2 3/8\,(4x-2x^2)\,dx = \underline{1/2}$$

Instructive Example

Let the continuous random variable X denote the current measured in a thin copper wire in miliamperes. Assume that the range of X is [4.9, 5.1] mA, and assume that the probability density function of X is

$$f(x) = 5, \qquad 4.9 \le x \le 5.1$$

Find the probability that a current measurement is less than 5mA.



$$P(X < 5\,mA) = \int_{4.9}^{5} 5\,dx = 1/2$$

$$\rightarrow 0.1 \times 5 = 0.5$$

Example-3.3

The lifetime in hours of a certain kind of radio tube is a random variable having a probability function given by

$$f_T(t) = \begin{cases} 0 & t \le 100 \\ 100/t^2 & t > 100 \end{cases}$$

P(getting 2 out of 5 which are broken in the first 150 hours)

$$= \binom{5}{2}\left(\tfrac{1}{3}\right)^2\left(\tfrac{2}{3}\right)^3$$

What is the probability that exactly 2 of 5 tubes in radio set will have to replaced within the first 150 hours of operation? Hint: Assume that the events E_i=1,2,3,4,5 that the ith tube wil have to be replaced within this time are independent.

T = lifetime of a component (Cont.)

Binomial

X = # of broken components in the first 150 hours (Discrete)

$\frac{1}{3}$ $\frac{1}{3}$ $\frac{2}{3}$ $\frac{2}{3}$ $\frac{2}{3}$

$$P(T \le 150) = \int_{100}^{150} 100/t^2\,dt = 1/3$$

$$\left(\binom{5}{2}\tfrac{1}{3}^2 \cdot \tfrac{2}{3}^3\right) \rightarrow \text{Binomial Distr.}$$